

A Multi-Cohort Decision-Evaluation Benchmark for EHR World-Model Components and Offline Reinforcement Learning

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ABSTRACT

Electronic health record (EHR) world-model components are increasingly connected to offline reinforcement learning (RL), yet comparisons can be driven by incompatible tasks, actions, rewards, support rules, or evaluator assumptions rather than algorithmic differences. We introduce a development benchmark that standardizes these layers across six ICU task constructions and releases complete results for six transition methods, five uncertainty-capable methods, 34 policy/planner labels, and nine off-policy evaluation (OPE) estimators. Across the transition inventory, no method is a global winner: the Transformer leads one-step error, whereas recursive point error, probabilistic sharpness, and calibration select the Gaussian ensemble, persistence/LOCF, or GRU-D depending on the metric and cohort. In heterogeneous exact-finite mechanisms, learned policies sometimes exceed matched fixed extremes, but model-free and world-model planners trade places across tasks and no learned method is uniformly best. Repeated-dataset OPE validates only null-response sanity contracts; no adaptive-response estimator contract passes the frozen joint gates. We therefore report no retrospective EHR policy value or winner. The contribution is an executable multi-cohort scaffold, a complete all-method performance inventory, and a failure decomposition from P/R/T component forecasting through policy-evaluation authorization.

CCS CONCEPTS

• **Applied computing** → **Health informatics**; • **Computing methodologies** → *Reinforcement learning*.

KEYWORDS

electronic health records, world models, offline reinforcement learning, benchmark, off-policy evaluation, uncertainty

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1 INTRODUCTION

EHR prediction benchmarks standardize cohorts, preprocessing, splits, and metrics [4, 14]. Offline RL over EHR trajectories adds a more fragile technical chain: recorded state must precede the action label, reward must follow it without mechanically encoding it, policy probability must remain inside empirical support, learned

dynamics must remain stable under recursion, and OPE must recover known values before retrospective scoring is authorized. Sepsis RL established a widely reused fluid-by-vasopressor action abstraction [6]; recent work connects latent EHR dynamics and imagination to policy learning [15]. These studies motivate a shared evaluator, but their different task and scoring contracts prevent direct algorithmic comparison.

Our primary contribution is the standardized comparison and failure decomposition across EHR P/R/T component learning, component model $\times$ planner combinations, model-based and model-free offline RL, known-value truth, retrospective support diagnostics, and cross-cohort evaluability. Transition accuracy is a validity layer rather than the endpoint. Temporal leakage control is a methodology requirement rather than the headline contribution.

Because no genuinely untouched role remains, all EHR findings are labeled *development benchmark evidence*. The benchmark evaluates P/R/T component and model-free offline-RL pipelines across heterogeneous EHR tasks, compares policy behavior against exact truth in constructed response mechanisms, and audits support, collapse, planning stability, and model exploitation on retrospective EHR data. The retrospective layer evaluates computational behavior, not policy value.

Contributions. We provide: (1) a common, time-ordered task scaffold with train-only fitting and subject-disjoint development roles; (2) an AI-Clinician-aligned K25 sepsis scaffold plus five task-specific extensions; (3) complete all-method transition, uncertainty, policy, and OPE inventories rather than winner-only summaries; (4) a heterogeneous exact-finite stress family that separates fixed-extreme dominance from adaptive policy discrimination; (5) a component-level bridge from prediction and uncertainty to known-value policy behavior and model exploitation; (6) repeated-dataset OPE coverage and authorization rather than estimator names or single-dataset interval inclusion; and (7) explicit non-execution when probability, support, or evaluator prerequisites fail.

Research questions. The benchmark asks: **RQ1**, how do P/R/T component families differ in one-step, recursive, reward, termination, uncertainty, and action-information fidelity? **RQ2**, how do model-based and model-free policies behave under interior-optimum, state-dependent, heterogeneous-response, and delayed-trade-off mechanisms? **RQ3**, when do learned models exhibit exploitation gaps, and which OPE contracts recover known policy values and ranks under repeated logged-data sampling? **RQ4**, which parts of this decision scaffold remain operational across heterogeneous cohorts without transferring sepsis-specific reward semantics?

2 RELATED DECISION BENCHMARKS

Table 1 separates task-specific retrospective EHR RL, EHR component/simulation models, and synthetic known-value benchmarks. It is representative rather than systematic: the purpose is to expose which evidence layers are connected under one contract, not to claim exhaustive coverage.

Prior work occupies distinct evidence surfaces. Real-EHR studies usually use task-specific retrospective evaluators; MedDreamer applies one method scaffold to two task-specific contracts; general EHR models may stop at trajectory simulation; and synthetic benchmarks supply known returns without validating real-EHR task construction. We connect six real-EHR task contracts to common P/R/T diagnostics, matched policy inventories, known-value evaluator tests, and explicit retrospective authorization boundaries. Our respiratory and shock tasks remain published-inspired adapters rather than reproductions of DeepVent or POPCORN. Healthcare offline-RL model selection remains sensitive to reward, support, and estimator choice [9, 13].

3 BENCHMARK SCAFFOLD

3.1 State, action, target, and roles

Each transition uses a fixed four-hour clock. Let

$$x_t = [v_t, m_t, \delta_t, c, a_{t-1}], \quad y_t = v_{t+1}, \quad (1)$$

where v_t is the pre-action value vector, m_t its observation mask, δ_t feature-wise recency, c static/pre-anchor context, and a_{t-1} the previous action abstraction. Current action, post-action measurements, future masks, post-outcome fields, and whole-stay diagnosis proxies are prohibited from x_t . The logged action a_t summarizes recorded exposure or settings over $[t, t + 4h)$; y_t begins at $t + 4h$. Every evaluated bin remains inside one ICU stay.

Throughout, $K = |\mathcal{A}|$ denotes the number of discrete action classes, not the number of observed transitions or treatment dimensions.

The interval label is deliberately called a *recorded-exposure action*: it can include adaptation during the interval and is not asserted to be fully assignable at its start. It is valid for logged-action-conditioned forecasting and retrospective policy-behavior diagnostics. In the known-value environments, by contrast, a_t is an assignable intervention in the constructed transition kernel. This distinction prevents the real-EHR label from inheriting the stronger semantics of the simulator action.

Imputation, normalization, action cutpoints, support masks, feature selection, behavior denominators, hyperparameters, and checkpoints are fitted or selected on development train/validation roles only. The roles are subject disjoint, but historical use of the available evaluation roles precludes calling any role a new confirmatory holdout. No row is used to claim a confirmatory best method.

3.2 AI-Clinician-aligned K25 sepsis scaffold

Sepsis is aligned to the AI Clinician lineage only at the four-hour fluid-quintile $\times$ vasopressor-quintile decision scaffold ($K = 25$). It uses local 33- and 40-variable views with masks/time gaps, prior action, and locally reconstructed SOFA/lactate reward components. It is not an exact reproduction: numerical reward

coefficients and parts of the source cohort implementation are not verifiable from the current artifact; the repaired anchor and stay-bounding rules intentionally differ; the local SOFA proxy can mechanically overlap vasopressor action; and anchor-relative 90-day outcome ascertainment is unavailable. We therefore retain an aligned scaffold and treatment-isolated reward sensitivity, not a canonical reproduction. The evaluated strict development subset has 3,440 episodes. A broader local K25 predecessor contains 27,236 episodes, but uses a window-overlap cohort rather than the repaired stay-contained construction; it is reported as scale context, not silently pooled with the current denominator.

The matched feature/reward/denominator comparison did not support choosing a preferred setting because terminal outcome availability and matched semi-synthetic reliability were incomplete. The K25 scaffold remains a compatibility surface despite weak factual action information.

3.3 Task-specific extensions

Table 2 summarizes extension roles and aggregate scale. An evaluated episode is one materialized ICU stay in the subject-disjoint development train or validation role. Earlier full-lineage counts are construction context only; their definitions and role coverage differ, so they are not alternative denominators for the reported results. Episodes should not be summed across cohorts because one person or stay may satisfy more than one task definition. Respiratory uses observed PEEP quintile $\times$ observed FiO₂ quintile ($K = 25$); a row is excluded when either setting is unavailable rather than mapped to no action. Its primary dense benchmark reward is

$$r_t^{\text{resp}} = \mathbb{I}[94 \leq \text{SpO}_{2,t+1} \leq 98] - .5\mathbb{I}[\text{SpO}_2 \text{ outside}] + \mathbb{I}[70 \leq \text{MBP}_{t+1} \leq 80] - .5\mathbb{I}[\text{MBP outside}], \quad (2)$$

with an SpO₂-only sensitivity. Shock uses fluid quintile $\times$ vasopressor quintile and two separated reward components,

$$r_t^{\text{MAP}} = \text{clip}((\text{MBP}_{t+1} - 65)/25, -1, 1), \quad (3)$$

$$r_t^{\text{lac}} = -\tanh(\text{lac}_{t+1} - \text{lac}_t).$$

All are frozen post-action benchmark objectives rather than transferred sepsis semantics. Sepsis world-model reward diagnostics use the lactate component; its reference terminal outcome is unavailable. AF/flutter remains world-model-only because strict action-reward ordering is unresolved. AKI and heart failure retain their original P/T transition surfaces and add separate decision-task extensions: severe AKI uses daily wait versus initiate RRT ($K = 2$, absorbing after initiation, at most three decisions), while heart failure uses a six-hour loop-diuretic-quartile by vaso/inotrope indicator action ($K = 8$, at most 12 decisions). Their post-action reward proxies define task-matched known-value mechanisms; they do not create retrospective EHR policy values.

4 METHODS COMPARED

4.1 P/R/T component models and metrics

The evaluated component is

$$\widehat{M} = (\widehat{P}_\theta(v_{t+1} \mid x_t, a_t), \widehat{R}, \widehat{T}), \quad (4)$$

not a complete generator of the future observation process. The core includes recurrent/GRU-D, causal Transformer, categorical

Table 1: Representative landscape of EHR decision-model evaluation. “Known return” means value supplied by the environment rather than estimated from retrospective trajectories. Multisite evaluation is separated from multiple decision-task contracts.

Study	Data and decision task	Model / policy layer	Evaluation evidence	Cross-task scope
AI Clinician [6]	MIMIC-III + eRI; sepsis fluid-vasopressor decisions	Finite MDP + policy iteration	Retrospective OPE and action-outcome concordance; no known return	One task; two datasets
POPCORN [2]	MIMIC-III acute hypotension; hourly K20 fluid-vasopressor actions	Prediction-constrained POMDP + PBVI	Forecasting/likelihood + CWPDIS, ESS, and support; no known return	One task
DeepVent [7]	MIMIC-III; four-hour ventilation settings	Model-free CQL	FQE and action-distribution diagnostics; no known return	One task
MedDreamer [15]	MIMIC sepsis + eICU ventilation	Latent RSSM + imagination-based policy learning	Learned-reward/relative-mortality metrics + CWPDIS/ESS; no known return	Two tasks; task-specific actions/rewards
EHR-MPC [12] (preprint)	Eight-hospital ICU data; sepsis policy evaluation	Generative EHR twin + MPC	Importance weighting + learned-twin simulation; no known return	One policy task; multiple sites
EHRWorld [11] (preprint)	MIMIC-IV longitudinal trajectories	Intervention-aware long-horizon model; no policy learner	Forecasting and simulation fidelity	No policy benchmark
EpiCare / DTR-Bench [3, 10]	Synthetic clinical decision tasks	Multiple offline and online RL methods	Simulator true return + OPE	Multiple synthetic tasks; no real-EHR task construction
Ours	MIMIC-IV; six real-EHR task contracts	Four P/R/T component families + matched policy layers	Adaptive exact-finite truth + monotone sensitivity + retrospective authorization boundaries; no EHR winner	Shared cross-task contract

Table 2: Episode counts and evaluation layers. “Evaluated” is the strict development train+validation denominator used for the reported component results. Earlier full lineages are context-only predecessor constructions, not pooled evaluation data. “Supported/K” gives train-supported EHR action classes over the discrete action space.

Cohort	Evaluated episodes	Earlier full lineage	Action contract	Benchmark role	Reward / value status
Sepsis	3,440	27,236	22/25 fluid-vasopressor	K25 sepsis scaffold	P/R/T plus support/collapse diagnostics
Respiratory support	21,679	–	12/25 PEEP-FiO ₂	Policy extension	P/R/T plus known-value stress tests
Shock	49,495	–	25/25 fluid-vasopressor	Policy extension	P/R/T plus known-value stress tests
AF/flutter	4,276	14,580	16/25 rate-rhythm	Component-only	P/T; action-reward ordering unresolved
AKI	21,083	–	P/T: K4 compact; policy: K2 wait/RRT	Decision extension	P/T plus known-value stress tests
Heart failure	3,986	32,552	P/T: K2 compact; policy: K8 diuretic × vaso/inotrope	Decision extension	P/T plus known-value stress tests

recurrent state-space, and Gaussian recurrent ensemble models; persistence and histogram gradient boosting are sanity controls. Neural runs use seeds 3408/3411/3414, hidden dimension 48, latent dimension 16 where applicable, batch size 512, learning rate 0.002, and weight decay 10^{-4} . Checkpoints minimize 0.7 RMSE+0.3 MAE on development validation after train-only fitting, with 8 minimum and 40 maximum epochs, patience 6, and one uniform 80-epoch extension when the frozen convergence trigger fires. The Gaussian ensemble combines the three recurrent members and is one derived row per task.

One-step RMSE/MAE uses only observed target cells after train-fitted normalization. Recursive rollout starts from the first validation state, feeds each predicted value back, marks generated future values as unobserved, increments recency, and follows the logged action sequence for eleven four-hour steps. At each horizon, error is scored against cells observed at that step in the recorded target mask. The scalar recursive error is the normalized observed-cell RMSE at the final executed horizon (step 11, 44 hours), averaged across available executed-seed rows for a method-cohort pair; all 924 horizon-resolved rows are released. Thus recursion tests value-state compounding under a fixed missingness convention; it does not claim to generate future measurement decisions. Reward heads are deterministic mappings from predicted state when available. Termination is evaluated natively, after train-only recalibration, and against a train empirical-hazard baseline.

Uncertainty reporting includes Gaussian NLL/CRPS, coverage at 50, 80, 90, and 95%, interval width and score, and risk-coverage.

Factual action information uses one fixed primary estimand throughout: shuffled-action minus observed-action RMSE on reward-relevant targets within matched strata, averaged over the four core model families and executed seeds. Each model-seed contrast also carries a 500-replicate materialized-episode bootstrap interval. It is a predictive-information diagnostic.

4.2 Planners and policies

The planner matrix uses exact supported-action search at horizon one and categorical cross-entropy method (CEM) planning at nominal horizons four and eight. Each CEM iteration evaluates 64 sequences, retains eight elites, updates categorical probabilities with smoothing 0.2, applies the frozen support mask at every step, and executes only the first action before replanning. Three iterations and matched model-query budgets are used. The effective horizon is truncated by the remaining task horizon; therefore AKI’s nominal H4 and H8 variants are both H3-effective and are retained only as budget-contract diagnostics. The uncertainty sensitivity subtracts 0.1 times predictive scale.

The model-free core includes behavior cloning, independent implementations of discrete BCQ [1] and CQL [8], a Soft-SPIBB conceptual adapter, a Decision Transformer contract adapter, and supported random, minimum, maximum, and severity-rule controls. Neural runs use three fixed seeds, one train/validation trial, at most 20,000 training windows, 12 epochs, hidden dimension 64, batch size 512, and discount 0.99. Fidelity labels distinguish independent

reimplementations from conceptual adapters. Deterministic controls are prespecified ϵ -soft distributions over the supported set with $\epsilon = 0.001$ for retrospective probability diagnostics; they are not scored by retrospective OPE.

4.3 Heterogeneous known-value and OPE evaluation

The primary policy surface crosses respiratory, shock, AKI-RRT, and heart-failure task profiles with six exact finite mechanisms: null response, a retained monotone control, interior optimum, state-dependent optimum, heterogeneous response, and delayed trade-off. Action counts, horizons, supported sets, behavior concentration, missingness, and termination prevalence are calibrated from aggregate EHR contracts; state dynamics, action response, toxicity, subtype, and counterfactual truth are specified synthetically. Dynamic programming supplies exact policy values. Hard gates require null-response equality, nonnegative exact regret, support compliance, categorical-CEM and receding-horizon execution, common-random-number precision, and complete inventory before method comparisons are read.

The four new mechanisms remove the fixed-extreme dominance built into the historical monotone control. They encode, respectively, an intermediate dose optimum, an optimum that varies with observed state, partially observed response heterogeneity, and delayed benefit versus immediate cost. An earlier composite adaptive family, in which a prespecified severity rule nearly recovers the exact policy, remains a separate construction sensitivity. Historical monotone matrices are retained only for evaluator regression testing.

All action responses, rewards, and termination mechanisms are known by construction. World-model architectures and policy methods are retrained within each environment; EHR-trained weights are not evaluated in these simulators. Five fixed training seeds compare five controls, five model-free methods, 24 world-model-planner combinations, and the exact dynamic-programming reference. Fixed-action controls are distinct from the state-dependent severity rule.

Exact dynamic-programming value is the primary estimand for every frozen policy. A separate common-random-number Monte Carlo check evaluates all policies from the same initial-state and exogenous-noise streams; policy randomness, training seeds, and evaluation seeds are separately recorded. Exact/Monte Carlo agreement and paired behavior-difference precision are hard evaluator checks, not substitutes for exact value. The complete policy-level release retains exact return, regret, behavior delta, Monte Carlo return and standard error, unsupported mass, training seed, and model-exploitation gap.

The OPE matrix tests trajectory IS and WIS; per-decision IS, WPDIS, and CWPDIS; DR and WDR; FQE; and support-restricted variants. A contract fixes estimator, behavior denominator, clipping, support rule, horizon, and regime. For each task-regime cell, 200 independent logged datasets are generated for respiratory, shock, and heart failure and 500 for AKI-RRT. Behavior denominators and nuisance models are refit inside each dataset, and each interval uses only that dataset. Empirical coverage is the fraction of independently generated datasets whose interval

contains the exact policy value; a 95% Wilson interval reports its sampling precision. Approval requires minimum policy-specific empirical coverage ≥ 0.90 ; Spearman rank and pairwise-order recovery ≥ 0.80 ; behavior-relative sign recovery ≥ 0.90 ; false-improvement probability ≤ 0.05 ; median ESS $\geq \max(100, 0.05N)$; zero unsupported target mass; and finite required values. The former single-dataset policy-set interval inclusion rate is retained as diagnostic-only and is not an authorization gate.

5 RESULTS

5.1 RQ1: P/R/T component fidelity

The full transition inventory comprises persistence/LOCF and HGB controls plus four world-model families. The Transformer has the lowest one-step RMSE on all six cohorts. At the final recursive horizon, the Gaussian ensemble leads five cohorts, while persistence/LOCF leads heart failure; when restricted to the four learned world-model families, the ensemble leads all six. Across the 24 core world-model-cohort bridge rows, normalized one-step error ranges from 0.669 to 0.773 and final-horizon error from 0.901 to 3.268. Thus the one-step selection reverses on every task. These are descriptive point leaders: neural methods have three seeds, the derived ensemble has one task-level row, and no subject-cluster leader interval is available.

Table 3 consolidates every executed comparison layer. It shows a stronger conclusion than a single aggregate leaderboard: some methods dominate one frozen metric, but the identity of the leader changes across point prediction, probabilistic recursion, adaptive policy comparison, and evaluator authorization. The complete layer-specific inventory and every granular result row are released.

Table 4 reports every transition method under the same task-balanced aggregation rather than only winners. Persistence/LOCF has no probabilistic output, and HGB has no matched action-information row; these missing cells are structural rather than suppressed results. The complete 36 cohort-method rows, including poor results, are released separately so users can reproduce a chosen cohort rather than reverse-engineer a headline leader.

Under the fixed action-information estimand, cohort means range from 0.0001 to 0.0023. Episode-cluster intervals exclude zero in all 10 model-seed rows for respiratory, shock, and AKI; 2/10 for AF/flutter; and 0/10 for sepsis and heart failure. Statistical separation can coexist with negligible magnitude and does not repair a missing repeated-action contract. The empirical-hazard termination baseline has lower Brier score than native and recalibrated learned heads in all 18 comparable K25-sepsis rows, so learned termination is a sensitivity rather than an unqualified planning component.

Uncertainty is also cohort- and metric-dependent. Across HGB and the four probabilistic world models, the one-step CRPS leader is the Transformer in five cohorts and the ensemble in sepsis. Under recursive rollout, the ensemble has the lowest CRPS in all six, whereas GRU-D has the lowest MACE in all six. The recursive 90% coverage range is especially broad in sepsis (0.623–0.941) and heart failure (0.657–0.934). Hence neither one-step point error nor one-step uncertainty identifies recursive sharpness and calibration jointly.

Table 3: End-to-end baseline and method atlas. CRPS, mean absolute coverage error (MACE), and exact regret are minimized. Point leaders are descriptive within a frozen layer and do not define a global cross-layer winner.

Layer	Metric	Evaluated inventory	Scope	Point-leader pattern
Transition	One-step RMSE	LOCF, HGB, GRU-D, Transformer, RSSM, ensemble (6)	6 EHR cohorts	Transformer 6/6
Transition	Final-horizon RMSE	Same six methods	6 cohorts; step 11 (44 h)	Ensemble 5/6; LOCF 1/6
Uncertainty	One-step CRPS / MACE	HGB plus four probabilistic WMs (5); LOCF unavailable	6 EHR cohorts	CRPS: Transformer 5/6, ensemble 1/6; MACE: Transformer 4/6, GRU-D 2/6
Uncertainty	Recursive CRPS / MACE	Same five probabilistic methods	6 EHR cohorts	CRPS: ensemble 6/6; MACE: GRU-D 6/6
Policy	Exact return / regret	5 controls + 5 model-free + 24 WM-planner labels (34)	24 heterogeneous exact-finite environments; 4,080 seed rows	Learned > both extremes in 11/16 new cells; no universal winner
OPE	Repeated-dataset authorization	IS, WIS, PDIS, WPDIS, CWPDIS, DR, WDR, FQE, support-restricted WPDIS (9)	2,200 independent datasets; 1,728 tuples	Null sanity only; adaptive 0/864; none retrospective

Table 4: Complete task-balanced transition/component performance. Each value first averages executed seeds within a cohort and method, then takes an unweighted mean over all six cohorts. RMSE, CRPS, and MACE are minimized; Cov90 is empirical recursive 90% interval coverage. Δ_a is matched-shuffle minus observed-action RMSE on reward-relevant targets and is predictive, not causal. Full cohort-specific rows are released.

Group	Method	1-step RMSE	Recursive RMSE	1-step CRPS	Recursive CRPS	Rec. Cov90	Rec. MACE	Δ_a
Control	Persistence/LOCF	0.777	1.095	–	–	–	–	–
Control	HGB residual	0.695	1.018	0.354	0.479	0.827	0.060	–
P/R/T	GRU-D	0.688	1.107	0.341	0.485	0.904	0.023	0.0011
P/R/T	Transformer	0.683	2.579	0.337	0.839	0.710	0.164	0.0010
P/R/T	Categorical RSSM	0.749	1.161	0.371	0.527	0.795	0.088	0.0019
P/R/T	Gaussian ensemble	0.684	0.969	0.338	0.458	0.937	0.056	0.0008

5.2 RQ2: Heterogeneous known-value policy performance

All 24 task-mechanism environments and 4,080 trained policy instances pass exact/Monte Carlo agreement and support checks; the release includes 2,880 world-model-planner cells and an empty execution-failure ledger. The frozen discrimination check asks whether any learned policy-seed row exceeds both matched minimum- and maximum-action controls. It passes in 11 of 16 prespecified new task-mechanism cells: 3/4 each for respiratory, shock, and heart failure, and 2/4 for AKI-RRT. Delayed trade-off remains fixed-control-led in all four profiles. Among the 11 successful cells, the best recorded learned row is model-free in eight and world-model-based in three. This cell-maximum check establishes that the mechanism family is not uniformly extreme-action dominated; it is not training-seed inference or a universal method ranking.

Figure 1 reports every one of the 34 non-oracle labels under the same five-seed aggregation in all 16 new cells. It confirms the benchmark-level conclusion: poor and strong baselines coexist, the leading learned family changes with task and mechanism, and no method dominates the matrix. The earlier composite adaptive family remains a useful sensitivity: its exact policy beats every fixed action, but the prespecified severity rule beats every learned method. Historical monotone matrices are retained only as evaluator smoke tests. All granular rows remain in the artifact rather than being filtered to winners.

5.3 RQ3: Exploitation and OPE disposition

The AI-Clinician-aligned K25 sepsis scaffold passes 98 of 1,800 horizon-specific pre-estimator gates. Material seed instability occurs in 26 of 50 rows, unsupported probability mass in 9 of 150, and marginal single-action collapse in 18 of 150. These failures

explain why probability-complete training output is not sufficient for policy-value scoring.

World-model planners also exhibit substantial simulator-to-truth disagreement. Across the 2,880 heterogeneous planner-seed cells, task-level mean absolute learned-model-versus-exact-return gaps range from 0.060 to 1.185, with a maximum of 1.826. These values remain on task-specific constructed reward scales and are not compared as cross-task performance; they show why a planner’s value inside its learned model cannot replace exact known truth.

Repeated-dataset evaluation generates 2,200 independent logged datasets and evaluates 1,728 frozen OPE-contract tuples. Table 5 shows the result. Forty tuples pass all gates, all in AKI-RRT null-response sanity cells; zero of 864 adaptive-response tuples passes. Adaptive failures arise across coverage, rank/order, sign, false-improvement, and ESS rather than a single implementation defect. The former fixed-dataset interval-inclusion result, including earlier task-matched AKI candidates, is retained in the artifact as diagnostic history but is not interpreted as frequentist coverage or retrospective authorization. No estimator is transported by family name and no retrospective EHR OPE value is reported.

5.4 RQ4: Cross-cohort evaluability

The common scaffold produces time-ordered transition tasks in all six cohorts, but higher evaluation layers differ. Respiratory and shock retain rich repeated actions, post-action reward proxies, P/R/T components, and matched known-value task profiles. AF/flutter has a supported $K = 25$ transition action but fails strict action-reward ordering and remains component-only. AKI and heart failure require separate decision contracts rather than promotion of their compact transition actions: daily absorbing RRT initiation and six-hour loop-diuretic by vaso/inotrope,

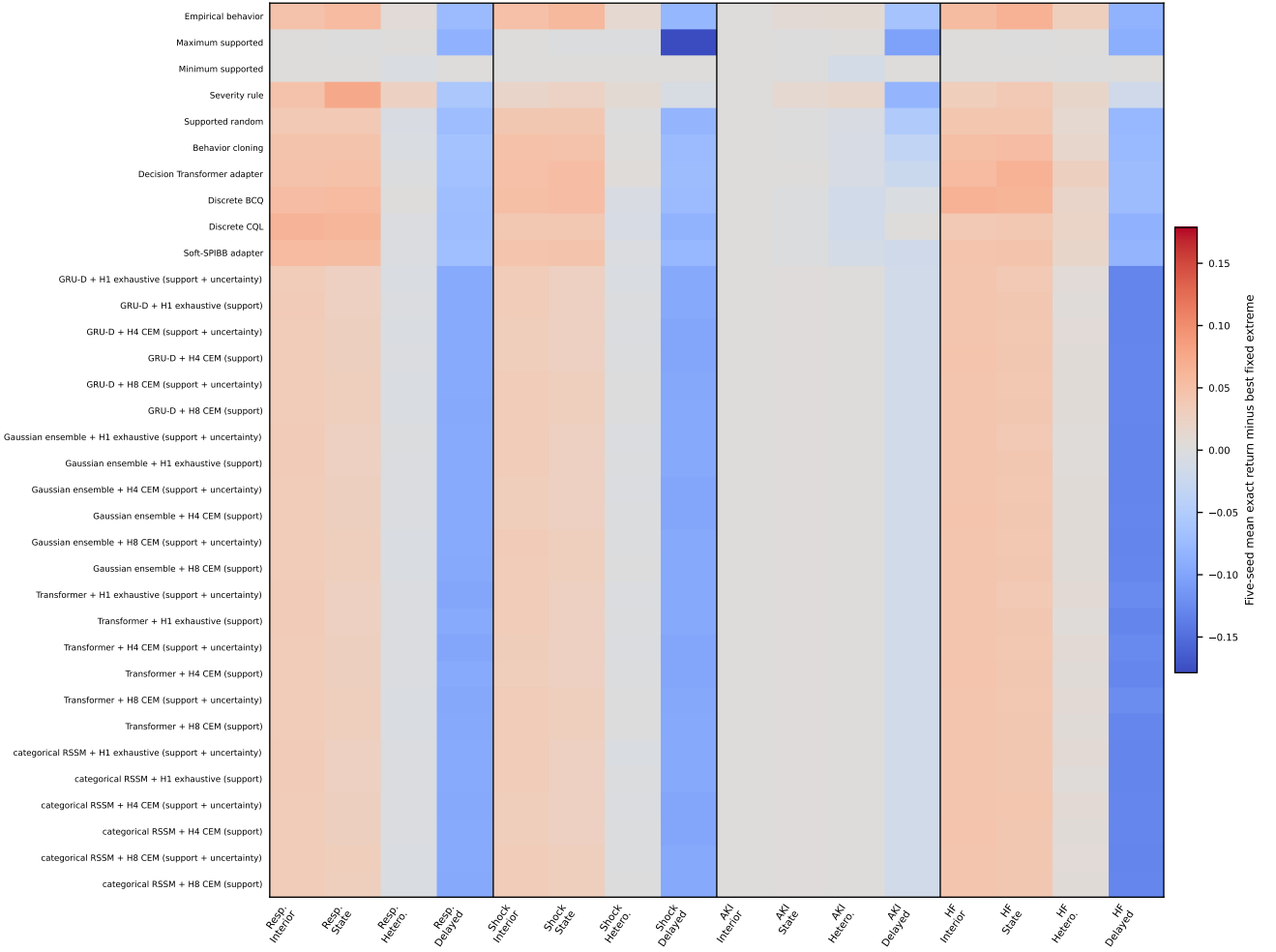


Figure 1: All 34 non-oracle policies across the 16 prespecified new task-mechanism cells. Color is five-seed mean exact return minus the better matched fixed extreme (red is higher; blue is lower). The frozen 11/16 discrimination decision uses the complete policy-seed cell maximum with a 10^{-12} tolerance; the heatmap is the seed-mean all-method inventory. Exact returns, regrets, support, seed variation, planner cells, and exploitation gaps are released.

Table 5: Repeated-dataset OPE gate pass counts. Each response regime has 864 frozen contract tuples. Coverage is empirical across independent logged datasets; all thresholds are frozen in Methods.

Gate	Adaptive response	Null response
Empirical coverage	64	56
Rank	266	864
Pairwise order	377	864
Improvement sign	121	864
False improvement	330	233
ESS	72	144
Support	864	864
Finite values	864	864
All gates	0	40

respectively. Both support constructed known-value model-based/model-free comparison. None yields an adaptive-response

repeated-dataset OPE contract that authorizes retrospective scoring. The benchmark therefore reports a six-task EHR component surface, four known-value policy profiles, one component-only task, and zero retrospective policy-value surfaces.

6 PREDICTION-TO-POLICY FAILURE DECOMPOSITION

Prediction quality is neither ignored nor treated as policy performance. An 80-row machine-readable contract matrix states which known-value properties come from EHR aggregates—action count/support, horizon, missingness, termination prevalence, and behavior concentration—and which are specified—state dynamics, action response, toxicity, subtype, and exact truth. World-model architectures are retrained independently in the known-value environments; EHR weights are not reused. Respiratory and shock

form the aligned bridge, while AKI and heart failure are explicitly labeled action-contract mismatch diagnostics.

The bridge is mixed rather than a stable prediction-to-policy law. In the aligned respiratory/shock subset, recursive RMSE versus exact policy regret has Spearman $\rho = 0.190$ ($n = 8$), uncertainty-calibration error versus planner seed dispersion has $\rho = 0.257$ ($n = 6$), and factual-action information versus policy discrimination has $\rho = 0.548$ ($n = 8$). Within-task family-rank concordance is -0.400 for respiratory and 1.000 for shock. These descriptive coefficients have small, dependent task-family samples and no confirmatory p-values. Together with the six-cohort one-step-to-recursive ranking reversal and heterogeneous exploitation gaps, they show that no single forecasting metric consistently predicts decision performance.

The resulting benchmark logic is compositional: a row may remain valid for P/R/T forecasting while failing planning discrimination, and a policy may remain valid for support/collapse analysis while failing value authorization. This prevents an upper-layer failure from erasing lower-layer evidence or a lower-layer success from silently authorizing a higher-layer claim.

7 REPRODUCIBILITY AND ACCESS

The reader package includes a benchmark card; task, feature, action, reward, planner, model, and estimator cards; sanitized policy-level and tuple-level result tables; a number-to-source audit; immutable source hashes; environment manifest; build commands; and credentialed reconstruction instructions. It exposes all 36 cohort-transition-method rows, task-balanced results for all six transition methods, all 4,080 heterogeneous policy-seed rows, all 2,880 world-model-planner cells, the earlier adaptive and monotone sensitivities, 10,368 policy-specific repeated-coverage rows, all 1,728 revised authorization tuples, horizon-resolved transition results, cohort-specific uncertainty summaries, and bridge coefficients rather than only headline aggregates. Code, contracts, synthetic fixtures, aggregate evidence, and clean-room commands are provided in the anonymous artifact at <https://anonymous.4open.science/r/ehrdyn-icu-65FB>. The source data are MIMIC-IV v3.1 [5]; access remains governed by PhysioNet credentialing and its data-use agreement. Raw EHR rows, identifiers, exact timestamps, trajectories, role-membership files, row-level predictions, tensors, and checkpoints are not redistributed.

Authorized users rebuild task arrays locally, fit all data-dependent transforms on development training roles, run development validation, execute exact known-value and repeated-dataset OPE evaluation, and regenerate the aggregate synthesis and PDF. Unrestricted users can validate the packaged contracts and aggregate hashes and run the known-value tests without MIMIC-IV. Independent credentialed reconstruction remains pending: it is a reproducibility check in which another MIMIC-credentialed user regenerates aggregate results from a clean clone, not external-data validation or a generalization experiment.

8 LIMITATIONS, DATA GOVERNANCE, AND SCOPE

The EHR findings are development benchmark evidence: no genuinely untouched confirmatory role is available. Component-model leader comparisons lack subject-cluster intervals, and the derived Gaussian ensemble contributes one row per task. Recursive evaluation predicts values under a fixed missingness convention rather than modeling a generative observation process; accordingly, we claim P/R/T components rather than a complete EHR world model with an observation head. The four-hour action is a recorded interval exposure, not guaranteed to be assignable at the bin start. Sepsis is an AI-Clinician-aligned K25 scaffold rather than an exact reproduction; its current 3,440-episode strict subset is much smaller than the 27,236-episode broader K25 predecessor and its reference terminal outcome is unavailable. AKI and heart-failure policy comparisons use separately constructed known-value tasks rather than the original EHR transition actions, and AF remains component-only.

The policy mechanisms are exact and heterogeneous but constructed; their results do not establish value on retrospective EHR trajectories. The frozen 11/16 discrimination statistic is a maximum over the complete learned policy-seed cell and therefore requires a seed-aware follow-up before it can support learned-family superiority. The repeated-dataset OPE audit covers null and composite-adaptive regimes, not all four new heterogeneous mechanisms, and authorizes no adaptive or retrospective scoring. MIMIC-IV is restricted, so the artifact shares code, contracts, synthetic fixtures, and aggregate results rather than row-level data. These are evidence boundaries of the technical benchmark.

9 CONCLUSION

We present a common decision-evaluation scaffold that connects EHR P/R/T component learning, recursive uncertainty, planning, model-free RL, heterogeneous known-value truth, and repeated-dataset OPE authorization. Its main empirical finding is not a universal winner: one-step, recursive, probabilistic, policy, and estimator layers produce different rankings and failure modes. By publishing exact contracts, every method row, fixed controls, and non-execution rules, the benchmark makes those differences reusable while preserving valid lower-layer results. Retrospective EHR policy ranking remains unavailable until an adaptive evaluator contract and policy-specific EHR gates both pass.

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